

NovaCart – Online Grocery Store

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Technology Stack: React.js (Frontend), Node.js/Express.js (Backend), MongoDB Atlas (Database), JWT Authentication, Mongoose ODM, Multer (File Upload)

i) Admin Sign-In – Authentication and Authorisation

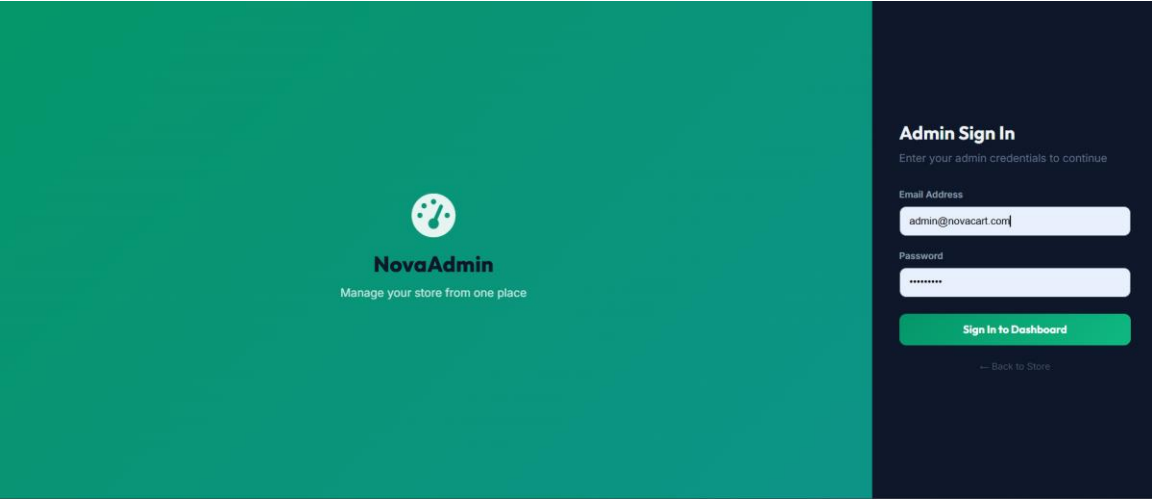
Authentication is the process of verifying the identity of a user, while Authorisation determines what resources or actions that verified user is permitted to access. In NovaCart, a token-based authentication strategy using JSON Web Tokens (JWT) is implemented to ensure secure admin access.

How It Works:

1. The admin submits their email and password through the Admin Login page (/admin/login).
2. The backend receives the credentials, searches for the user in MongoDB, and compares the stored hashed password with the submitted password using bcrypt.
3. If the credentials are valid and the user role is 'admin', a signed JWT token is generated with a 7-day expiry and returned to the client.
4. The frontend stores this token in localStorage and attaches it to all future HTTP requests in the Authorization header as a Bearer token.
5. Protected routes pass through 'protect' middleware to verify the JWT and 'adminOnly' middleware to ensure the user role is admin.

Key Concepts:

- Hashing – Passwords are never stored as plain text. bcrypt hashing ensures security.
- JWT – A stateless token that stores user identity and role information.
- Middleware – Express middleware intercepts requests and applies security checks before controllers execute.
- Role-Based Access Control – Only admin-role users can access admin dashboard routes.



NovaAdmin

Dashboard

Products

Orders

Users

NovaCart Admin Administrator

Back to Store

Sign Out

Welcome back, NovaCart Admin

Dashboard Overview

Total Revenue
Rs. 400

Total Orders
1

Customers
3

Products
2

Recent Orders

Order	Customer	Total	Status	Date
#F7E0CB	John Customer	Rs. 400	Confirmed	3/31/2026

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User Management

3 Users

User	Email	Role	Joined	Actions
John Customer	john@test.com	User	3/30/2026	Make Admin
NovaCart Admin	admin@novacart.com	Admin	3/30/2026	Remove Admin
TestUser	testxyz@test.com	User	3/30/2026	Make Admin

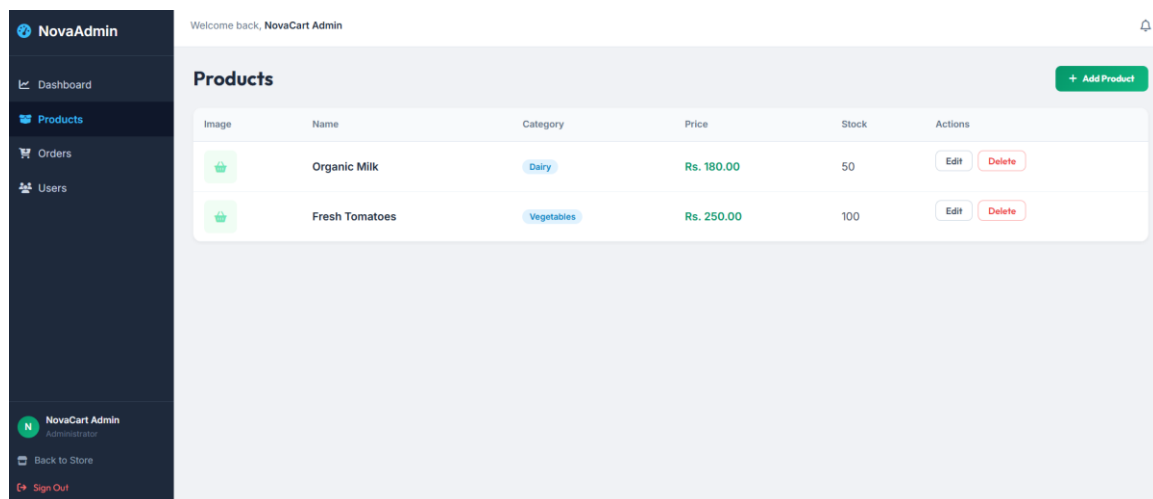
ii) Product Management (CRUD) – Dashboard to Frontend Synchronisation

CRUD represents Create, Read, Update, and Delete operations that manage application data. In NovaCart, the Admin Dashboard allows administrators to manage products stored in the MongoDB database. All changes are immediately reflected on the customer storefront because both systems use the same database.

- Create – Admin adds a new product through the dashboard form. The data is sent to POST /api/products and saved in MongoDB.
- Read – The shop page retrieves products using GET /api/products and displays them as product cards with search and category filters.
- Update – Admin edits a product and sends a PUT /api/products/:id request which updates the product document in MongoDB.
- Delete – Admin deletes a product using DELETE /api/products/:id which removes the product from the database.

Product images are uploaded using Multer middleware and stored in the server uploads directory. The image path is saved in MongoDB and served as a static URL so the frontend can display the image using an HTML image tag.

Since the admin dashboard and storefront both interact directly with MongoDB, the database acts as a single source of truth, ensuring all updates are visible immediately.



The image shows two screenshots of the NovaAdmin dashboard. The top screenshot displays the 'Edit Product' form, which is divided into two main sections: 'Basic Information' and 'SEO Optimization'. The 'Basic Information' section includes fields for Product Name (Organic Milk), Category (Dairy), Price (Rs.) (180), Stock Quantity (50), and a Description (Fresh organic whole milk). The 'SEO Optimization' section includes fields for URL Slug (organic-milk), Meta Title (Buy Organic Milk | NovaCart), and Meta Description (Order fresh organic milk online). The bottom screenshot displays the 'Orders Management' table, which lists orders with columns for Order ID, Customer, Items, Total, Date, Status, and Update. The table shows one order with ID #96F7E0CB, customer John Customer, 1 item, total Rs. 400.00, date 3/31/2026, and status Confirmed.

Edit Product

Basic Information

Product Name *
Organic Milk

Category *
Dairy

Price (Rs.) *
180

Stock Quantity *
50

Description
Fresh organic whole milk

Product Image
Choose File No file chosen

SEO Optimization

URL Slug *
organic-milk
URL-friendly identifier (no spaces, use hyphens)

Meta Title
Buy Organic Milk | NovaCart

Meta Description
Order fresh organic milk online

Orders Management

Order ID	Customer	Items	Total	Date	Status	Update
#96F7E0CB	John Customer john@west.com	1 item(s)	Rs. 400.00	3/31/2026	Confirmed	Confirmed

iii) SEO Tags, Description and Meta Details on Product Page

Search Engine Optimisation (SEO) improves a website's visibility on search engines. For an e-commerce store, each product page should have unique metadata to help search engines understand the content.

- Title Tag – The main headline shown in search results.
- Meta Description – A summary of the page that appears under the title in search results.
- Meta Keywords – Keywords describing the product.
- URL Slug – A readable URL that includes keywords relevant to the product.

The product schema includes fields such as description, slug, metaTitle, metaDescription, and metaKeywords. The admin enters these values through the dashboard.

The react-helmet-async library dynamically inserts these values into the HTML head section, ensuring each product page has unique SEO metadata and search-friendly URLs.

iv) Cart and Checkout – Sandbox Deployment

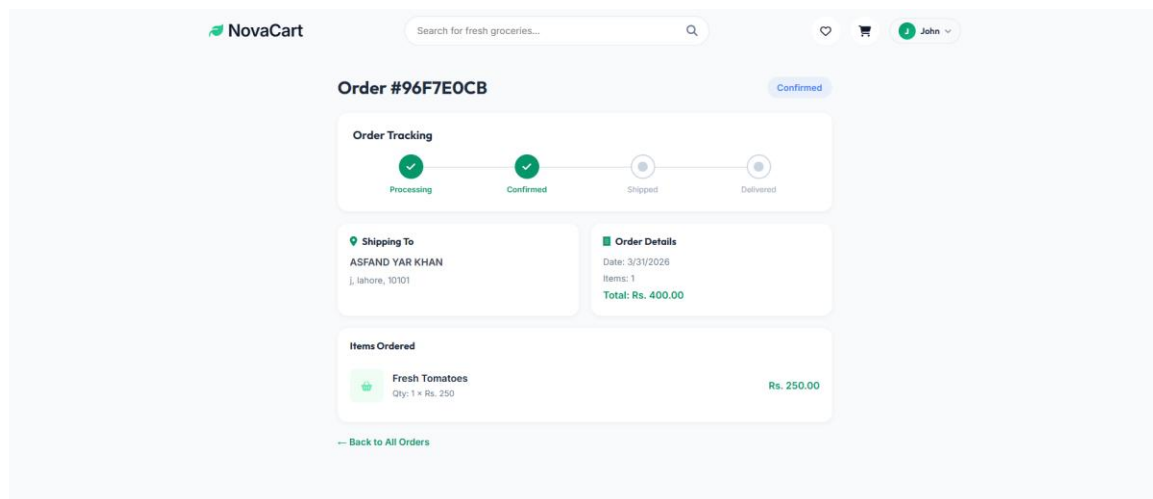
The shopping cart and checkout system manage the purchasing workflow in NovaCart. The cart is implemented using React Context which provides global access to cart data across components.

- Logged-in Users – Cart data is stored in MongoDB within the user document.
- Guest Users – Cart data is stored in localStorage so it persists after page refresh.

Cart operations include adding items, removing items, updating quantity, and clearing the cart after order completion.

During checkout, users provide shipping information and confirm their order. The system currently supports Cash on Delivery as a sandbox payment method, allowing full checkout functionality without real payment gateways.

When the order is placed, the system stores order details in MongoDB and clears the cart. The user is then redirected to the order confirmation page.



v) User Management – Sign In, Sign Up, Cart, Wishlist and Order Tracking

User management controls how customers register, log in, manage carts, maintain wishlists, and track orders.

- Sign Up – Users register with name, email, and password. Passwords are hashed using bcrypt before storage.
- Sign In – Users log in with email and password and receive a JWT token for authentication.
- Cart – Logged-in users have their cart stored in the database while guests use localStorage.
- Wishlist – Users can save products and later move them directly to the cart.

- Order Management – Orders store product details, totals, shipping information, and status.
- Order Tracking – Users can monitor order progress from processing to delivery.

Protected routes ensure that only authenticated users can access pages such as checkout, orders, and wishlist. If no valid token is found, the system redirects the user to the login page.

